MY SQL ASSIGNMENT 2

```
DROP DATABASE IF EXISTS employee;
CREATE DATABASE employee;
USE employee;

CREATE TABLE departments (
  department_id INT PRIMARY KEY,
  department_name VARCHAR(100) NOT NULL
);

CREATE TABLE location (
  location_id INT PRIMARY KEY AUTO_INCREMENT,
  location VARCHAR(100) NOT NULL
);

CREATE TABLE employees (
  employee_id INT PRIMARY KEY,
  employee_name VARCHAR(50),
  gender ENUM('M','F'),
  age INT,
  date_of_joining DATE,
  designation VARCHAR(255),
  department_id INT,
  location_id INT,
  salary DECIMAL(10,2),
  email VARCHAR(100),
  FOREIGN KEY (department_id) REFERENCES departments(department_id),
  FOREIGN KEY (location_id) REFERENCES location(location_id)
);
```

```
INSERT INTO departments (department_id, department_name) VALUES
(1, 'Software Development'),
(2, 'Marketing'),
(3, 'Data Science'),
(4, 'Human Resources'),
(5, 'Product Management'),
(6, 'Content Creation'),
(7, 'Finance'),
(8, 'Design'),
(9, 'Research and Development'),
(10, 'Customer Support'),
(11, 'Business Development'),
(12, 'IT'),
(13, 'Operations');

INSERT INTO location (location) VALUES
('Chennai'),
('Bangalore'),
('Hyderabad'),
('Pune');

INSERT INTO employees (employee_id, employee_name, gender, age, date_of_joining, designation, department_id, location_id, salary) VALUES
(5001, 'Vihaan Singh', 'M', 27, '2015-01-20', 'Data Analyst', 3, 4, 60000),
(5002, 'Reyansh Singh', 'M', 31, '2015-03-10', 'Network Engineer', 12, 1, 80000),
(5003, 'Aaradhya Iyer', 'F', 26, '2015-05-20', 'Customer Support Executive', 10, 2, 45000),
(5004, 'Kiara Malhotra', 'F', 29, '2015-07-05', NULL, 8, 3, 70000),
(5005, 'Anvi Chaudhary', 'F', 25, '2015-09-11', 'Business Development Executive', 11, 1, 55000),
(5006, 'Dhruv Shetty', 'M', 28, '2015-11-20', 'UI Developer', 8, 2, 65000),
(5007, 'Anushka Singh', 'F', 32, '2016-01-15', 'Marketing Manager', 2, 3, 90000),
(5008, 'Diya Jha', 'F', 27, '2016-03-05', 'Graphic Designer', 8, 4, 70000),
```

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(5009, 'Kiaan Desai', 'M', 30, '2016-05-20', 'Sales Executive', 11, 3, 55000),
(5010, 'Atharv Yadav', 'M', 29, '2016-07-10', 'Systems Administrator', 12, 4, 80000),
(5011, 'Saanvi Patel', 'F', 28, '2016-09-20', 'Marketing Analyst', 2, 1, 60000),
(5012, 'Myra Verma', 'F', 26, '2016-11-05', 'Operations Manager', 13, 2, 95000),
(5013, 'Arnav Rao', 'M', 33, '2017-01-20', 'Customer Success Manager', 10, 3, 75000),
(5014, 'Vihaan Mohan', 'M', 30, '2017-03-10', 'Supply Chain Analyst', 10, 2, 60000),
(5015, 'Ishaan Kumar', 'M', 27, '2017-05-20', 'Financial Analyst', 7, 1, 85000),
(5016, 'Zoya Khan', 'F', 31, '2017-07-05', 'Legal Counsel', 4, 4, 100000),
(5017, 'Kabir Nair', 'M', 28, '2017-09-11', 'IT Support Specialist', 12, 2, 80000),
(5018, 'Ishan Mishra', 'M', 25, '2017-11-20', 'Research Scientist', 9, 3, 75000),
(5019, 'Ishika Patel', 'F', 29, '2018-01-15', 'Talent Acquisition Specialist', 4, 4, 55000),
(5020, 'Aarav Nair', 'M', 32, '2018-03-05', 'Software Engineer', 1, 1, 90000),
(5021, 'Advik Kapoor', 'M', 26, '2018-05-20', 'Finance Analyst', 7, 3, 85000),
(5022, 'Aadhya Iyengar', 'F', 28, '2018-07-10', 'HR Specialist', 4, 4, 60000),
(5023, 'Anika Paul', 'F', 30, '2018-09-20', 'Public Relations Specialist', 2, 2, 70000),
(5024, 'Aryan Shetty', 'M', 27, '2018-11-05', 'Product Manager', 5, 1, 95000),
(5025, 'Avni Iyengar', 'F', 31, '2019-01-20', 'Data Scientist', 3, 4, 100000),
(5026, 'Vivaan Singh', 'M', 29, '2019-03-10', 'Business Analyst', 3, 2, 75000),
(5027, 'Ananya Paul', 'F', 32, '2019-05-20', 'Content Writer', 6, 3, 60000),
(5028, 'Anaya Kapoor', 'F', 26, '2019-07-05', 'Event Coordinator', 6, 1, 60000),
(5029, 'Arjun Kumar', 'M', 33, '2019-09-11', 'Quality Assurance Analyst', 12, 2, 80000),
(5030, 'Sara Iyer', 'F', 28, '2019-11-20', 'Project Manager', 5, 1, 90000);

SELECT DISTINCT salary FROM employees;
SELECT age AS Employee_Age FROM employees;
SELECT salary AS Employee_Salary FROM employees;

SELECT * FROM employees
WHERE salary > 50000 AND date_of_joining < '2016-01-01';

SELECT * FROM employees WHERE designation IS NULL;
```

```
UPDATE employees
SET designation = 'Data Scientist'
WHERE employee_id = 5004;

SELECT * FROM employees
ORDER BY department_id ASC, salary DESC;

SELECT * FROM employees
WHERE YEAR (date_of_joining) = 2018
ORDER BY date_of_joining
LIMIT 5;

SELECT SUM(e.salary) AS finance_total
FROM employees e
JOIN departments d ON e.department_id = d.department_id
WHERE d.department_name = 'Finance';

SELECT MIN(age) AS min_age FROM employees;

SELECT l.location, MAX(e.salary) AS max_salary
FROM employees e
JOIN location l ON e.location_id = l.location_id
GROUP BY l.location_id, l.location;

SELECT designation, AVG(salary) AS avg_salary
FROM employees
WHERE designation LIKE '%Analyst%'
GROUP BY designation;

SELECT d.department_name, COUNT(e.employee_id) AS emp_count
FROM departments d
```

```
LEFT JOIN employees e ON d.department_id = e.department_id
GROUP BY d.department_id, d.department_name
HAVING COUNT(e.employee_id) < 3;

SELECT l.location, AVG(e.age) AS avg_age
FROM employees e
JOIN location l ON e.location_id = l.location_id
WHERE e.gender = 'F'
GROUP BY l.location_id, l.location
HAVING AVG(e.age) < 30;

SELECT e.employee_name, e.designation, d.department_name
FROM employees e
INNER JOIN departments d ON e.department_id = d.department_id;

SELECT d.department_name, COUNT(e.employee_id) AS total_employees
FROM departments d
LEFT JOIN employees e ON d.department_id = e.department_id
GROUP BY d.department_id, d.department_name;

SELECT d.department_name, COUNT(e.employee_id) AS total_employees
FROM departments d
LEFT JOIN employees e ON d.department_id = e.department_id
GROUP BY d.department_id, d.department_name;

SELECT l.location, e.employee_name
FROM employees e
RIGHT JOIN location l ON e.location_id = l.location_id;
```